

Training Configuration and Summary:

Sequencing:

- timesteps: 60
- batch_size: 128

Model Hyperparameters:

- min_units: 128
- max_units: 256
- units_step: 64
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh']
- optimizers: ['adam', 'Nadam']

Tuning:

- max_epochs: 30
- hyperband_iterations: 3

Directories:

- directory: my_dir
- project_name: simplernn_hyperparameter_tuning

SimpleRNN Training Report

Report Generated: 2025-04-09_20-41-43

Training Start: 2025-04-09 20:31:03

Training End: 2025-04-09 20:39:31

Training Duration: 508.57 seconds

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: Nadam

Dropout Rate: 0.20

Units per Layer:

Layer 1: 128 units, Layer 2: 192 units

Performance Metrics:

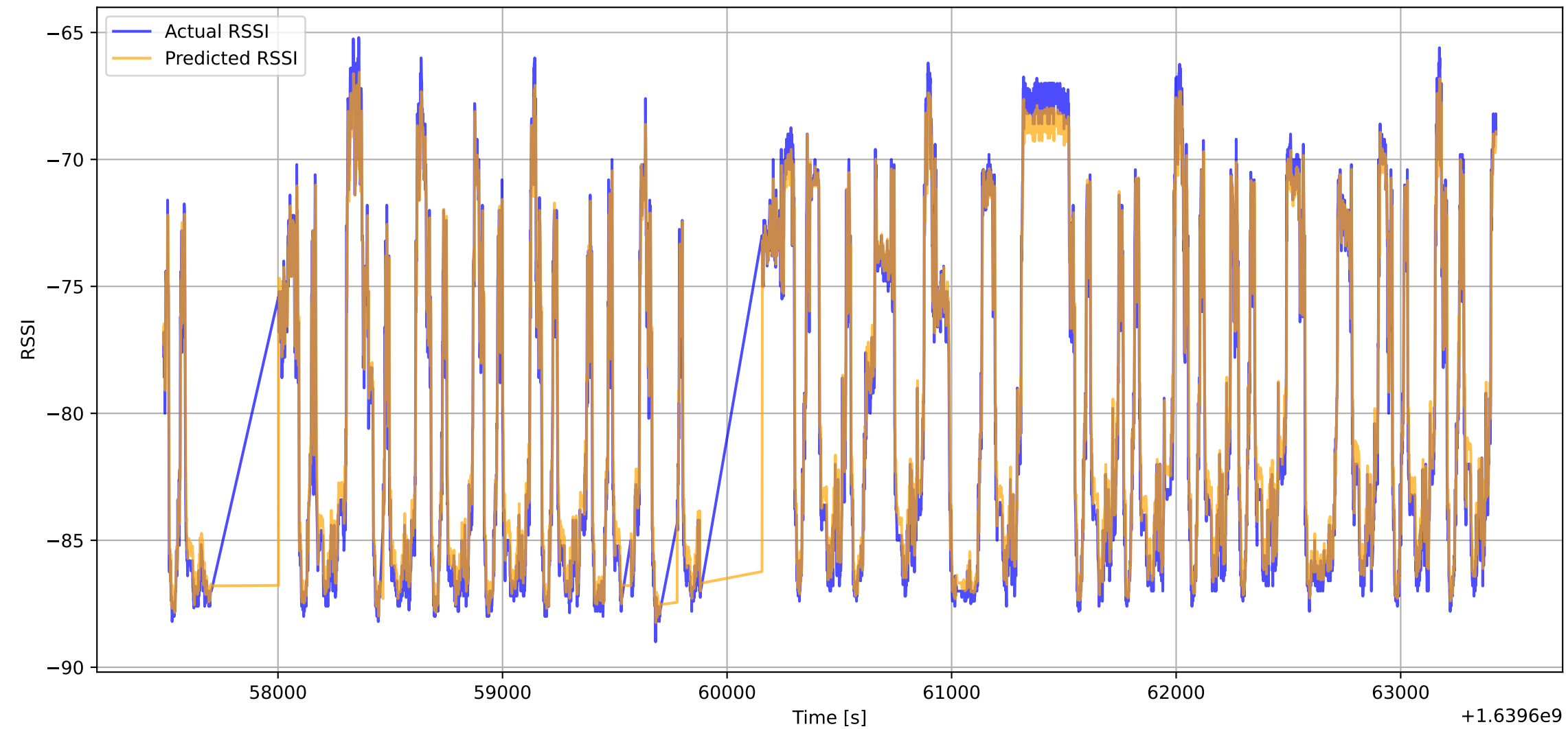
MAE: 0.9780

MSE: 1.8717

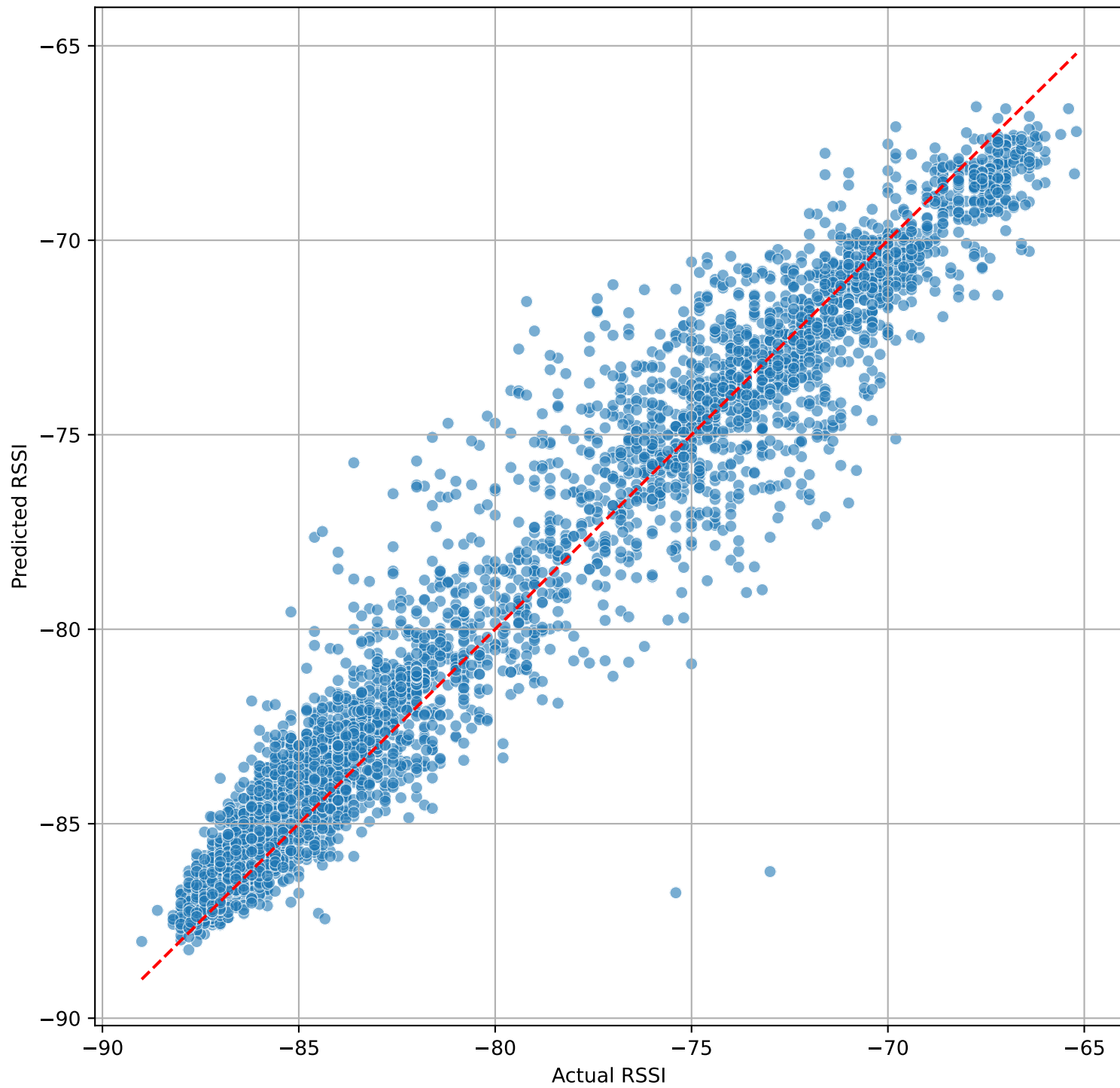
RMSE: 1.3681

R2 Score: 0.9575

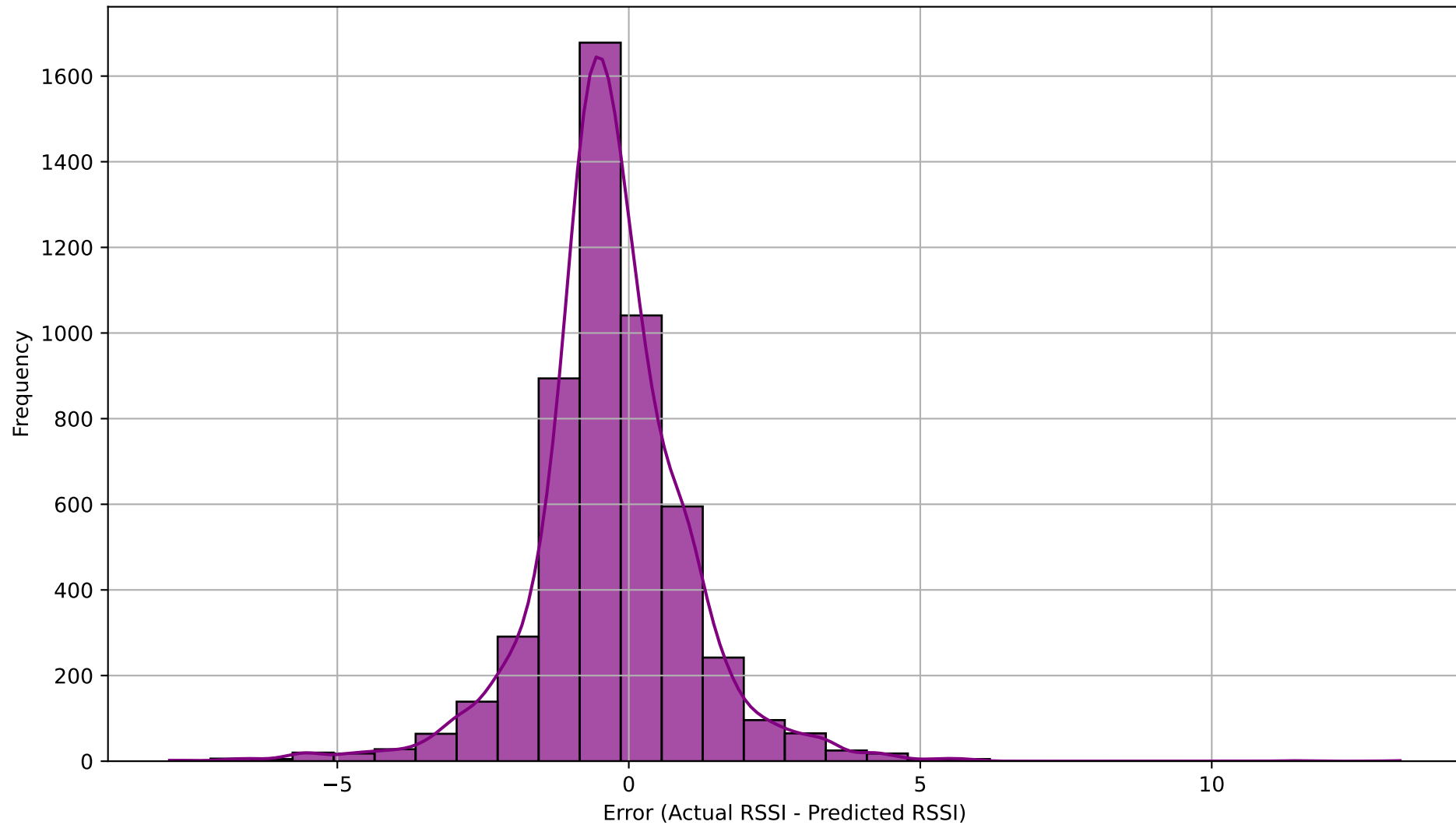
Actual vs Predicted RSSI Over Time



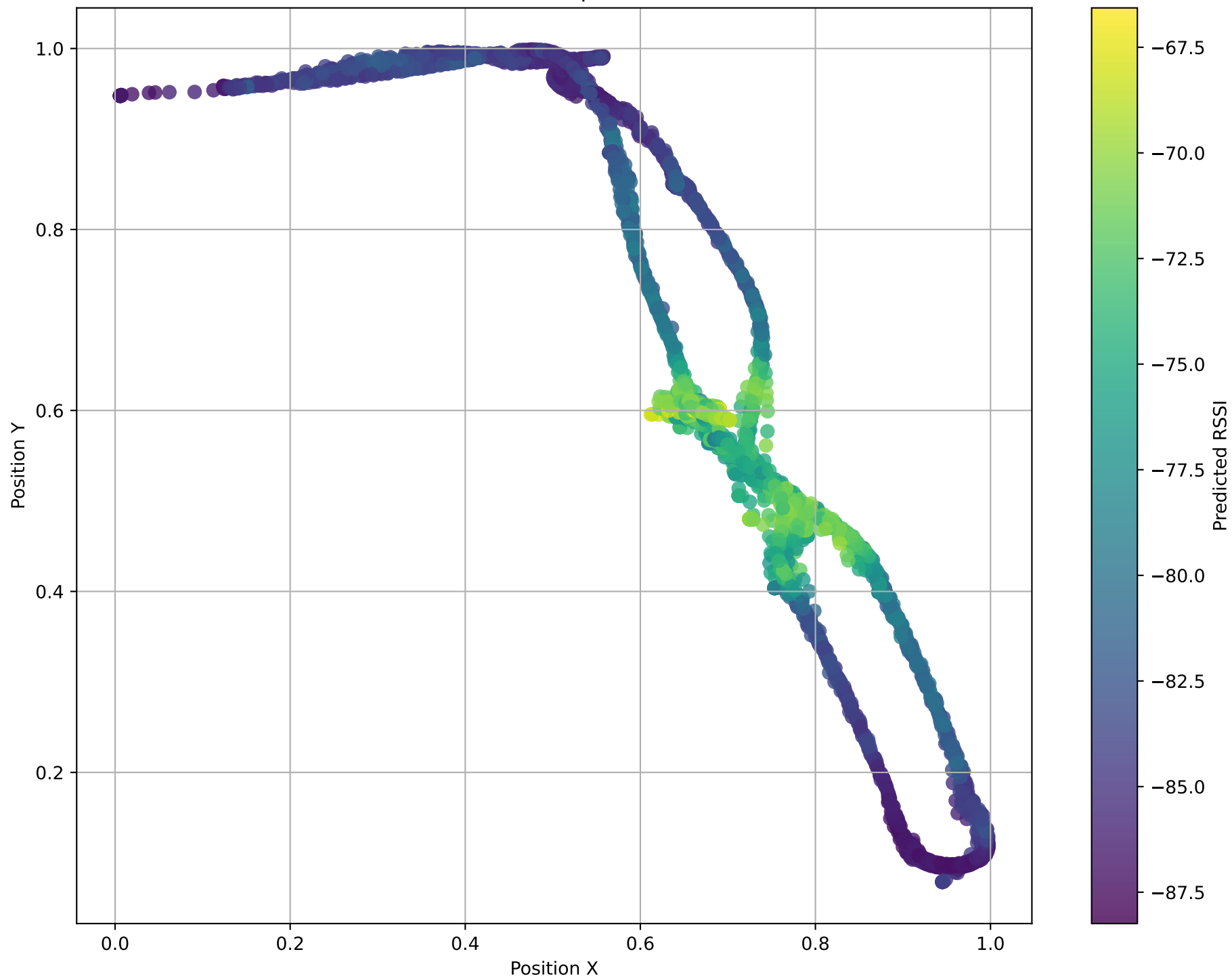
Actual vs Predicted RSSI



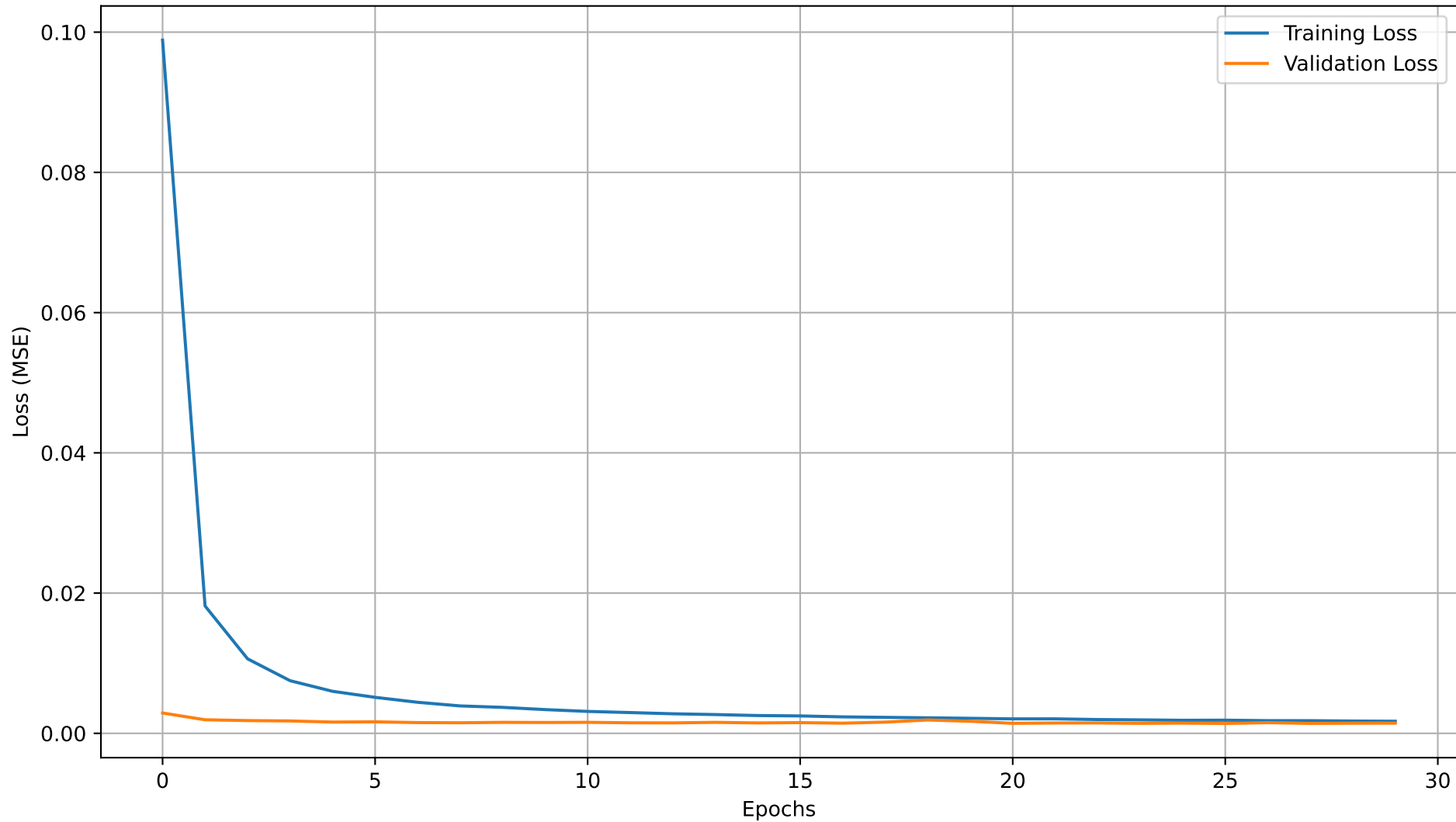
Distribution of Prediction Errors



Predicted RSSI in Spatial Context



Training and Validation Loss Over Epochs



Training and Validation MAE Over Epochs

